

Regression Models and Regularization Techniques

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1 Introduction

Regression is a supervised machine learning technique used to predict continuous numerical values.

2 Linear Regression

Linear Regression assumes a linear relationship between the input variable and the output variable.

$$Y = b_0 + b_1X$$

where:

- Y = predicted value
- X = input variable
- b_0 = intercept
- b_1 = slope

2.1 Example

Study Hours	Score
1	50
2	60
3	70
4	80

The model is:

$$Score = 40 + 10 \times Hours$$

For 5 hours:

$$Score = 40 + 10(5) = 90$$

3 Polynomial Regression

Polynomial Regression can model curved relationships.

$$Y = b_0 + b_1X + b_2X^2$$

Example:

$$Price = 22000 - 1000X - 300X^2$$

For $X = 4$:

$$Price = 22000 - 1000(4) - 300(16)$$

$$Price = 13200$$

4 Ridge Regression

Ridge Regression adds an L2 penalty:

$$Cost = MSE + \lambda \sum b_j^2$$

Example:

$$10^2 + 8^2 = 164$$

5 Lasso Regression

Lasso Regression adds an L1 penalty:

$$Cost = MSE + \lambda \sum |b_j|$$

Example:

$$|10| + |8| + |6| = 24$$

Lasso can remove features by setting coefficients to zero.

6 Elastic Net Regression

Elastic Net combines Ridge and Lasso.

$$Cost = MSE + \lambda_1 \sum |b_j| + \lambda_2 \sum b_j^2$$

7 Mean Squared Error (MSE)

The Mean Squared Error is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Example:

Actual values:

$$[100, 200, 300]$$

Predicted values:

$$[110, 190, 280]$$

Errors:

$$(-10)^2 + (10)^2 + (20)^2$$

$$100 + 100 + 400 = 600$$

$$MSE = \frac{600}{3} = 200$$

8 Comparison of Regression Models

- **Linear Regression**

- Main Idea: Straight-line relationship
- Best Visualization: Straight line

- **Polynomial Regression**

- Main Idea: Curved relationship
- Best Visualization: Parabola

- **Ridge Regression**

- Main Idea: Handles noisy data
- Best Visualization: Smoothed line

- **Lasso Regression**

- Main Idea: Removes useless features
- Best Visualization: Coefficients become zero

- **Elastic Net**

- Main Idea: Combines Ridge and Lasso regularization
- Best Visualization: Mixed behavior

9 Conclusion

Linear Regression is simple and effective for linear data.

Polynomial Regression handles non-linear relationships.

Ridge reduces large coefficients.

Lasso performs feature selection.

Elastic Net combines Ridge and Lasso.

MSE is a common metric used to evaluate regression models.